MetaLearnHub: System Architecture & Technical Documentation

An Open-Set Few-Shot Meta-Learning Visual Classification System
Government College of Engineering, Nagpur | Dept of CSE | Group 07

1. Executive Summary & Problem Formulation

Conventional deep convolutional networks rely on massive datasets and closed-set fixed assumptions. When applied to few-shot novel classes, standard models suffer from severe overfitting, high retraining latency, and false-positive overconfidence. MetaLearnHub solves this by implementing an episodic meta-learning framework. Rather than learning fixed boundaries, the network constructs an inductive metric space that adapts instantaneously from 1-5 support images without neural network retraining.

- **Transferable Feature Learning:** Generic feature extraction across diverse animal categories.
- **Zero-Leakage Assurance:** Atomic group-aware splitting for paired clean and augmented files.
- **Instant Adaptation:** Prototype computation in under 25 milliseconds without backpropagation.
- **3-Gate Open-Set Rejection:** Distinguishes known categories from out-of-distribution inputs.

2. System Architecture

Layer	Technologies	Key Responsibilities
Presentation Tier	React 18, TypeScript, Vite, Tailwind CSS	Dashboard, dataset audit, live training monitor, evaluation lab, class registration, query interface.
Controller Tier	FastAPI, Python 3.10+, BackgroundTasks	REST endpoints, worker thread orchestration, file uploads, JSON streaming.
ML Engine	PyTorch 2.x, Torchvision, Scikit-learn	ResNet-18/50 backbones, 4 meta-learning algorithms, episodic samplers, LOCO evaluator.
Persistence	File System, JSON, PyTorch .pt tensors	Class registry (data/class_registry/), checkpoints/, experiments/ logs.

3. Data Engineering & Zero-Leakage Protocol

Paired clean and augmented images (e.g., Ayrshirecattle100.jpg and Ayrshirecattle100_c.jpg) risk crossing split boundaries. MetaLearnHub parses filenames into a unified sample_group_id and guarantees atomic group allocation. The Leave-One-Class-Out (LOCO) protocol completely excludes one breed from meta-training to benchmark genuine adaptation.

4. Deep Backbones & Meta-Learning Algorithms

Visual backbones include ResNet-18 (512-dim embedding) for fast episodic sweeps and ResNet-50 (2048-dim) for high-capacity open-set classification, projected onto the unit hyper-sphere via L2-normalization. The platform unifies four algorithms:

- **Prototypical Networks (ProtoNet):** Computes mean class centroids c_k; classifies via negative distance softmax.
- **Matching Networks (MatchingNet):** Evaluates query against every individual support exemplar via softmax attention.
- **Relation Networks (RelationNet):** Replaces static distance with a 4-layer learnable non-linear Relation Module.
- **MAML (Optimization-Based):** Prototype-guided linear head adapted via 5 SGD gradient steps in <20 ms.
- **Ensemble Fusion:** Weighted combination: 35% ProtoNet + 30% MatchingNet + 20% RelationNet + 15% MAML.

5. Conventional Baseline vs. Few-Shot Meta-Learning

Evaluation Dimension	Conventional Supervised Baseline	MetaLearnHub Meta-Learning
Data Efficiency	Requires 500+ images; overfits on K=5	Adapts reliably from 1 to 5 images.
Adaptation Latency	1,500 - 5,000 ms (full gradient backprop)	12 - 45 ms (instant prototype computation).
Generalization on K=5	100% train accuracy; <35% test accuracy	75% - 85% accuracy on unseen test classes.
Open-Set Rejection	Closed-set; forces false positive on unknown	3-Gate rejection flags out-of-distribution inputs.

6. Dynamic Class Registry & 3-Gate Open-Set Engine

Classes and prototypes are persisted in data/class_registry//. Pre-flight batch audit calculates pairwise coherence %, flags visual outliers, and warns of collisions. Unknown query inputs must satisfy all three gates:

- **Gate 1 (Distance Gate):** Distance to prototype must be within class radius r_k = max(mean + 1.25*std, 1.05*max_dist, floor).
- **Gate 2 (Margin Gate):** Best candidate must separate from second-best competitor by relative margin >= 0.005.
- **Gate 3 (Confidence Gate):** Softmax confidence must be >= 50%; otherwise rejects as MATCH NOT FOUND.

7. Execution & Operational Commands

- **run.bat:** Starts FastAPI backend (port 8000) and React frontend (port 5173) simultaneously.
- **python run_few_shot_cli.py:** Runs headless automated or interactive few-shot CLI testing.
- **python init_base_classes.py:** Seeds 10 base animal categories into persistent class registry.